PharmaNutrition 5 (2017) 35–46

Contents lists available at ScienceDirect

PharmaNutrition

journal homepage: www.elsevier.com/locate/phanu

Mushroom nutraceuticals for improved nutrition and better human health: A review

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ARTICLE INFO

Article history:
Received 13 January 2017
Accepted 1 February 2017
Available online 9 February 2017

Keywords:
Mushrooms
Antioxidants
Nutraceuticals
Bioactive molecules
Nutritional and therapeutic role

ABSTRACT

Mushrooms have always been treasured and appreciated due to their immense role in curing various degenerative diseases. Also their unique taste and flavour make them a demanding food for every man's plate. Information on their chemical composition, nutritional value and therapeutic properties has expanded dynamically during the last few years. Mushrooms contain 50 to 65% total carbohydrate, 19 to 35% proteins (with various biological and medicinal activities such as lectins) and 2 to 6% fat content of their dry matter. In mushrooms unsaturated fatty acids are found to be predominant over saturated fatty acids especially palmitic acid, oleic acid and linoleic acid, while the proportion of linolenic fatty acids is very limited. Mushrooms rich in fat soluble vitamins along with ergosterol content are thought to be the only vegetarian source for vitamin D. Data on fibre content and composition is limited but polysaccharides like β-glucans boost up the nutritional value of mushrooms. It has been observed and established that mushrooms are wonderful food with their immense nutraceutical properties. And for this one of the reasons is the presence of bioactive molecules i.e. β-glucans, triterpenoids, antioxidants etc. These molecules are not only therapeutically strong but also immune-modulators. In this paper all the bioactive molecules in mushrooms along with their possible mechanisms have been discussed and highlighted so that these can be better harnessed for the mankind.

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http://dx.doi.org/10.1016/j.phanu.2017.02.001
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1. Introduction

Human relationship with mushrooms is fascinating since they have been used both as a food and medicine for last few decades. The studies on Cultural Revolution reveal that mushrooms were an integral part of the human diet since the time when they were food gatherer and hunters [1]. But now the usage of mushrooms has expanded up to a wider extent not only as food but also in the area of pharmaceuticals, nutraceuticals and cosmeceuticals for the mankind. According to current estimates, mushrooms constitute at least 12,000 species worldwide and out of that 2000 species are reported as edible. About 35 edible mushroom species are commercially cultivated whereas nearly 200 wild species used for medicinal purposes [2]. *Agaricus bisporus* is the most cultivated mushroom worldwide followed by *Lentinus edodes* and *Pleurotus* spp. [3].

Many researchers have documented that edible mushrooms are source of variety of nutraceutical compounds such as polysaccharides (β-glucans), dietary fibres, terpenes, peptides, glycoproteins, alcohols, mineral elements, unsaturated fatty acids, antioxidants like phenolic compounds, tocopherols, ascorbic acid etc [4]. Also the presence of specific bioactive compounds makes these mushrooms therapeutically valuable right from strengthening the immune system till the cure and prevention of life threatening diseases like heart diseases, hypertension, cerebral stroke and cancers. Mushrooms are also known to exhibit antifungal, anti inflammatory, antitumor, antiviral, antibacterial, hepatoprotective, anti diabetic, hypolipedemic, antithrombotic and hypotensive activities [5]. The therapeutic and neutraceutical exploration of mushrooms gives the impression that they are the next generation food, not only in providing quality protein but also in curing deadly diseases like cancers, tumors and nervous disorders. There is a huge challenge to better exploit this wonderful gift of nature and bring it forward from the farm to fork level.

2. Nutritional components of mushrooms

The total carbohydrate content in the fruiting bodies of mushrooms accounts for 50 to 65% on dry weight basis. It comprises of sugars which are monosaccharides, their derivatives and oligosaccharides. The carbohydrates contain some amount of alcoholic sugar, such as mannitol and trehalose. Trehalose is known to synthesize stress-responsive factors in human cells when exposed to environmental stresses like heat, cold, oxidation, desiccation etc by retaining cellular integrity. The possible mechanism behind this is the prevention from protein denaturation which usually degrades under stress conditions [6]. Mushrooms rich in proteins contain all the essential amino acids and of these the rich ones are glutamic acid, aspartic acid and arginine. Besides these, the two unusual amino acids γ-amino butyric acid (GABA) a non essential amino acid and ornithine which are known for their peculiar physiological activities have also been found [7].

Mushrooms contain unsaturated fatty acids with total lipid (crude fat) content 20–30 g kg^{-1} DM. They are found to be rich in linoleic and oleic acids [8]. Linoleic acid has been reported to exhibit anti-carcinogenic effects on almost all stages of tumorigenesis in animal models of breast, prostate, and colon cancers and also in reducing the tumor cell growth by altering the 5-lipoxygenase metabolite, 5-hydroxyeicosatetraenoic acid (5-HETE) and expressing the five-lipoxygenase activating protein (FLAP) [9]. Linoleic acid is also a precursor of 1-octen-3-ol (an alcoholic group of mushrooms) which is the key aromatic compound contributing to mushroom flavour [10]. The lipid fraction of mushrooms contains tocopherol an important antioxidant component. Mushrooms are good reservoir of vitamins specially vit. B complex and vit. D. Interestingly mushrooms are the only non animal based food that contains vit. D [11–13]. Recent studies have indicated that mushrooms when exposed to UV light under certain conditions produces vit. D_2 in amounts greatly higher than that that of daily requirements of vitamin D [14–16]. The process of vit D_2 formation takes place through a higher than that of daily requirements photochemical reaction in which fungal sterol, ergosterol, is converted to vitamin D_2 through a series of photochemical and thermal reaction catalyzed by ultraviolet (UV) radiation coming from sunlight [17].

Mushrooms are rich in potassium, calcium, phosphorus and magnesium. Sodium is relatively less in mushrooms and hence they are thought to be a good option amongst other vegetables for the hypertensive people [18]. Fresh mushrooms with soluble and non- soluble fibers are proved to lower the total cholesterol levels and thereby good in managing cardiovascular diseases [19]. The presence of dietary fiber (DF) with non dietary carbohydrates (NDCs) including β - glucans, polysaccharides-protein Complexes (PSPC), chitin, hemicelluloses, mannans, xylans and galactose provides a wide range of health benefits to the humans. Mushrooms also contains large number volatile compounds which are inbuilt with some enzyme system that are capable of catabolising aromatic substrates, the major one's are 3-octone, 3-octanol, 1-octen 3-ol, benzaldehyde, octanol, and 2-octen-1-ol (Table 1).

2.1. Bioavailability of nutrients

Certain diseased conditions require specific nutrient management. Bioavailability refers to the nutrient absorption and utilization in the body and is affected by many factors. Mushrooms are excellent functional foods and contain selenium, vitamin D_2, ergothioneine, Vitamin B_1 and Iron etc. It is interesting to mention that the bioavailability of any nutrient depends on the mushroom variety [28]. Mushroom contains 0.4–2.0 mg/g (dry weight) of ergothioneine (sulfur containing amino acid) [29]. Various researchers have reported good bio availability of ergothioneine

Table 1
The comparative information about the proximate composition (g/Kg) and energy value (kcal/Kg) of some edible mushrooms.

Species	Dry matter	Crude protein	Ash	Lipid	Carbohydrates	Energy	References
Agaricus bisporus(white)	87.3	140.8	97.4	21.8	740.0	325	Kalac [20]
Agaricus bisporus (brown)	83.6	154.3	113.6	16.7	715.4	303	Kalac [20]
Pleurotus eryngii	110.0	110.0	61.8	14.5	813.7	421	Kalac [20]
P. ostreatus	106.0	284.0	86.0	47	354.0	–	Ahmadetal. [21]
Lentinus Edodes	202.2	228	60	21	644	411	Bisen et al. [22]
Volvariella volvaceae	90.3	280	10	33	503	305	Mashandete & Cuff [23]
Auricularia auricular	94	360	52	43.7	285	–	Usha & Sugune [24]
Flammulina velutipes	121.3	176.0	74.3	28.9	731	378	Reis et al. [25]
Calocybe gambosa	90.8	154.6	138.9	83	698.2	317	Vaz et al. [26]
Calocybe indica		214	131	49.5	485	320	Alam et al. [27]

in human subjects. Calvo et al. conducted an interventional study on 37 pre-diabetics subjects and reported consumption of 100 g of fresh mushroom for 16 weeks resulted in increased ergothioneine content (317 ng/mL at baseline to 677 ng/mL) along with increased ORAC (oxygen radical absorption capacity), anti-inflammatory hormone (adiponectin) and significant decreases in serum oxidative stress inducing factors like carboxymethyllysine (CML) and methylglyoxal (MG) [30]. 16 g of button mushroom powder has also been found to significantly increase the ergothioneine content in RBCs [31]. Other mushroom species rich in ergothioneine are king oyster, oyster, maitake, and shiitake [32].

Mushrooms are rich in dietary selenium and are known to reduce oxidative stress. *Boletus edulis* has the highest selenium content (approx 20 μg Se/g dry weight). Many animal model studies have been reported highlighting the bioavailability of selenium. In a study conducted by Da Silva et al. feeding selenium fortified *P.ostreatus* mushroom for 5 weeks resulted in increase in the total plasma Se of 64 wistar rats [33]. Another study by Maseko et al. demonstrated significant boost in the GPx-1 and GPx-2 gene expression and colon GPx-1 enzyme activity when fed with selenium-enriched button mushrooms [34]. The bioavailability of Se is dependent on its chemical form of Se present in it and the same are bound to the polysaccharides in the cell wall. It is due to this reason that selenium bound to chitin might not be available to the humans [35]. Apart from this, mushrooms contain ergosterol which gets converted to Vit. D_2 under UV light exposure. Studies on 26 healthy adults having low 25(OH) D levels were conducted for 2 weeks and were fed with the UV treated mushroom soup. It was found that the levels of 25(OH) D were significantly higher than the control and it was quite comparable with the vitamin D_2 supplement [36,37]. Similar studies have been conducted by Stephenson et al. where higher levels of 25 (OH) D have been observed after six weeks of supplementation indicating the strong absorption in the serum [38]. The uptake of minerals by the body indicated potential uses of mushrooms as food source, dietary supplements or as neutraceuticals. Another study reports that administration of UVB exposed mushrooms also enhances bone growth and mineralization. Some studies also emphasize on the effects of processing on the vit. D_2 content in mushrooms. Report by Bogusz et al. suggests that ingestion of dried white button mushroom was equally effective in increasing the serum 25 (OH) D levels as the commercial supplements [39]. Mehrotra et al. reported low absorption of vit. D_2 in prediabetic subjects upon feeding with cooked UVB treated mushrooms as compared to the fresh ones [40].

Other nutrients like vit. B_{12} in mushrooms is similar to the ones found in beef, liver, and fish, suggesting its high bioavailability and can be important for individuals consuming a vegan diet over a lifetime [41]. Supplementation of products with dried mushroom powders like shiitake [42] and *pleurotus* [43] have confirmed to increase the blood hemoglobin concentration and liver Fe levels in vivo models.

2.2. Antinutritional factors

There are various types of antinutritional factors which are present in food and affect the human health. They impair the intake, uptake or utilization of other food components thereby decreasing the bioavailability of the nutrients causing discomfort and stress. Some of the major anti nutritional factors present are tannins, phytates, hemagluttinins, oxalates, saponins, protease inhibitors etc [44]. Since mushrooms are consumed in both raw and cooked form, the study of risks involved due to these antinutritional factors is important. Certain hemagluttinins found in *A.bisporus* and *P.ostreatus* have shown to interfere with the protein absorption in-vitro models and led to the formation of lesions in small intestine [45]. However, it is noteworthy to mention that few studies have reported safer limits of anti nutritional factors in mushrooms. Lower contents of phytates as compared to green leafy vegetables have been reported in *A. bisporos* [46,45]. Content of phytates were found to be in range of 0.57 mg/100 g to 1.09 mg/100 g indicating the safe consumption of *Agaricus* in raw form. Similarly, tannins in *Ganoderma* were found to be very low i.e. 2.29 −0.24 mg/gm [47]. Oly and Obiakor, carried out studies on three mushrooms namely button, oyster and Crimini mushroom and found it in the range of 0.77-1.55 mg/100 g [48]. The levels were over ten times below the safe limit as prescribed by WHO [49]. Comparable phytate results have been reported by Ogbe and Obeka, (2.43 mg/100 g) in *Ganoderma* mushroom [50] and 0.5–0.12% in Nigerian mushrooms by Afiukwe et al. [51]

Table 2
Effects of processing on nutritional composition of mushroom.

Species	Method of storage	Incubation period	Nutrients affected	Reference
A. bisporus	Stored at 12 °C	12 days	Reduced sugar content mannitol and fructose Total free amino acids increased from 77.92 to 140.57 g kg^{-1}	Tseng and Mau [54]
A. bisporus	Blanching at 95–100 °C	15 min	Reduced mineral levels	Coskuner and ozdemir [57]
A. bisporus	Freezing at −25 °C Canning and salting	6months	Reduction in protein by 24.3% 22.2% 16.54% in freezing, canning and salting respectively. Reduction in free amino acids (tyrosin, alanine, glutamine, cysteine) and flavor 5′-nucleotides in all treatments.	Liu et al. [53]
P. ostreatus	Microwave processing Frying	– –	Reduced Fe, Zn, Mn, Ca and Cu Increased Iron content	Ziarati & Ghasemynezhad [58]
P. ostreatus	Freezer-storage	12months	Decreased levels amino acids (alanine, glycine, serine, histidine, methoinine and threo nine)	Jaworska et al. [52]
P. ostreatus	Oven dried at 60 °C Blanched at 88 °C Brining (25% brine solution)	Till the constant weight obtained For 1 min. 30–60 min	Reduction in protein content and enhanced carbohydrate content Reduction in protein, fat content and enhanced carbohydrate content Remarkable reduction in protein, fat and carbohydrate content	Mayunja et al. [59]
A. bisporus and *B. edulis*	Canning Blanching in low-methylated pectin before canning	12 months	Reduction in amino acids by 2–16% in *A. bisporus* and 3–24% in *B. edulis.* The contents were better as compared to simple canning by 9–16% *A. bisporus* and 22–27% *B. edulis.*	Jaworska et al. [60]

3. Effects of storage on nutritional components of mushrooms

Due to the high percentage of moisture in mushrooms their processing is a necessary step to make it reachable. Like other vegetables they can be baked, fried pickled, canned, dried etc. but in India most of the population like to consume vegetables either fresh or with little or no processing. So as per the requirement mushrooms can be dried, frozen or canned. Blanching, steeping in solutions like L-ascorbic acid, sodium metabiosulphate etc are other some methods that can be done to preserve fresh mushrooms. Based on the scientific reports (given in Table 2) it has been observed that all the methods in one or other way affect the nutritional composition of the mushrooms while storage. Jowarska et al. says that soluble nitrogenous substances can be leached out into the brine solution during canning which can leads to the reduction in amino acid content 2–16% in *A. bisporus* and 3–24% in *B. edulis* mushroom species [52]. Freezing is the best method for the preservation, *A.bisporus* preserved for 6 months resulted in better nutrients restoration in freezing as compared to salting and canning. It was found that *A. bisporus* had the lowest protein content (16.5%) in salting and the flavor 5′-nucleotides were also high in freezing (6.22 mg/g) as compared to canning (2.83 mg/g) and salting (1.71 mg/g) [53]. Whereas storing at an ambient temperature 12 °C in PVC packets for 12 days *A. bisporus* found to increase the total free amino acids as well as improve the flavor 5′-nucleotides [54]. Flavour of mushroom products has been seen to be significantly affected by the thermal processing [55]. The non-volatile compounds of *A. bisporus* have been found to be greatly affected by canning procedure [56]. A study done on mushroom soups made up of *A. bisporus* indicated that the flavours and free amino acids were more restored in microwave cooked soup as compared to the boiled and autoclaved methods [55].

4. Bioactive nutraceutically important compounds

Mushrooms nutraceuticals have been used since ancient times for treating and curing thousands of human ailments. Their nutritional and culinary characteristics attracted the researchers, pharmacists, nutritionists to haul out their usage for the common people and to be included them as one of the vegetable under food kingdom. Its excellent medicinal power such as vit. B (panthothenic acid) important to build up the nervous system, mineral like selenium and ergothioneine an natural antioxidant to protect human cells from damage and the β-glucans (Fig. 1) marked for strengthening the immune system by inducing both the adaptive and innate immune responses makes these mushrooms a wonderful key source for making nutraceuticals. Moreover the presence of wonder molecules like polysaccharides, low molecular weight proteins, terpenes, glycoprotein and bioactive compounds again reinforce the usage of such magic food for the better mankind. The current review is discussing such active compounds in detail.

4.1. Polysaccharides

Mushroom polysaccharides have already been isolated and characterized like pleuran from *Pleurotus* species, lentinan and erothionine in *L. edodes*, ganoderan from *Ganoderma lucidium* and agaritine from *Agaricus* and calocyban from *Calocybe indica* [61]. All the mushroom polysaccharide contains a common β-linked glucose backbone but the pattern and degree of branching vary species to species. However, some polysaccharides having galactose or mannose as their backbone have also been found in mushrooms. These polysaccharide are known to possess various physiological activities like antitumor activity, immunomodulatory action, antioxidant activity, antiviral activity, anti-inflammatory actions and anti-carcinogenic (given in Table 3). The antitumor action of polysaccharides is mediated through a thymus-dependent immune mechanism (Fig. 2), which involves the activation of cytotoxic macrophages, monocytes, neutrophiles, natural killer cells, dendritic cells, and chemical messengers (cytokines, such as interleukins, interferons, and colonystimulating factors) which triggers the complementary and acute phase responses [62].

Mushroom polysaccharide extracted from *A. bisporus* exhibits excellent inhibiting action against human breast cancer [64]. Also proteogylcans from *A. blazei* are known to have strong immunomodulatory properties which are therapeutically important in controlling cancers and immune deficient diseases through the upregulation of dendritic cells maturation [65]. Recently a noval hetropolysaccharide composed of glucose units isolated from *P. eryngii* showed a great antitumor activity against HepG-2 cells [66]. The poysaccahride-protein complex isolated from aqueous extract of *Phellinus rimosus* found to have profound effects towards radiation-induced oxidative stress [67]. Interestingly the anti fatigue activities has also been observed in the polysaccharides extract of *Hericium erinaceus*, which extend the utilization of these kinds of novel polysaccharides for sports nutrition [68]. *Cordyceps sobolifera* also found to prosper in medicinally important polysaccharides and polysaccharide extract can be used to protect and treat the body from renal injuries [69].

4.2. Bioactive protein molecules

Proteins with biological functions show physiological activity in the gastrointestinal tract by enhancing nutrient absorption, inhibiting enzymes, and modulating the immune system to defend against pathogens [71]. Mushroom bioactive proteins and peptides such as lectins, fungal immunomodulatory proteins, ribosome inactivating proteins, antimicrobial proteins, ribonucleases, and laccases are an important part of functional components with great value of pharmaceutical potential [72]. Lectins are the non immune proteins or glycoproteins binding specifically to cell surface carbohydrates and have been studied for their antiproliferative, antitumor, and immunomodulatory activities [73]. Some other proteins which exhibit enzymatic activities are ribosomal

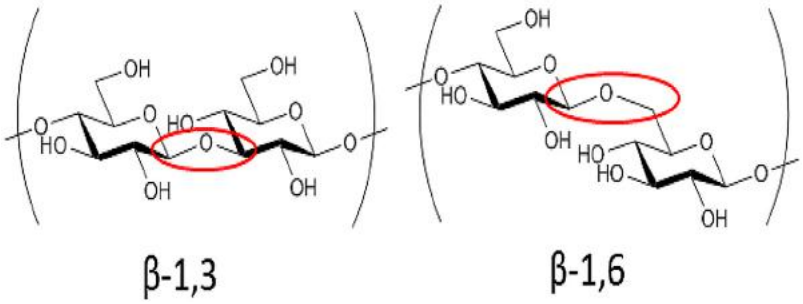

Fig. 1. (a & b): Structure of β-glucan linkages.

Table 3
Important mushroom polysaccharides and their health benefits [70].

Mushroom	Type of polysaccharides	Health benefits
Agaricus bisporus	Heteropolysaccharides	Activation of macrophages
Agaricus bitorquis	Homopolysaccharides	Activation of natural killer cells
Agaricus blazei	Glucan-protein complex	Activation of T lymphocytes
Auricularia auricula-judae	Homopolysaccharides	Anti-viral activity
Boletus erythropus	Homopolysaccharides	Antimicrobial activity
Calocybe indica	Homopolysaccharides	regulate lipogenesis
Ganoderma lucidum	Heteropolysaccharides	Induction of apoptosis
Geastrum saccatum	Glucan-protein complex	Treatment in stomach cancer
Grifola frondosa	Heteroploysaccharides Grifloan	Antitumor activity
Lentinus edodes	Heteropolysaccharides Lentinan	Antitumor activity
Phellinus linteus	Homopolysaccharides	Increase production of interleukin
Pleurotus eryngii	Homopolysaccharides	Antiproliferative effect
P. florida	Homopolysaccharides	Inhibit tumoral cell to cell adhesion
P. ostreatus	Homopolysaccharides	Increase gastrointestinal motility
Poria cocos	β-glucans type polysaccharides	Treatment of colon cancer
Polyporus rhinocerus	β-glucans type polysaccharides	Treatment of colon cancer
Schizophyllum commune	Homopolysaccharides Schizophyllan	Antitumor activity
Sparassis crispa	Homopolysaccharides	Lipid peroxidation inhibition
Termitomyces eurhizus	Homopolysaccharides	Anti-aging effects
T. microcarpus	Homopolysaccharides	Hepatoprotectiveactivity

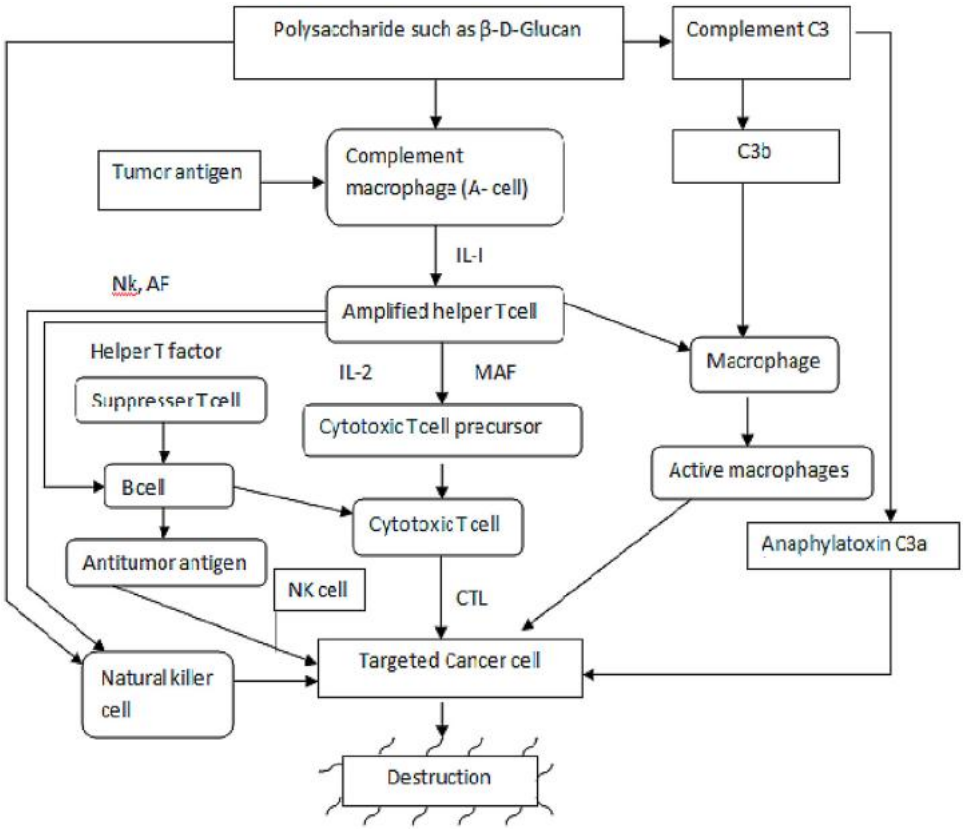

Fig. 2. Anticancer mechanism of mushroom polysaccharides [63].

inactivating proteins (RIPs), laccase, fungal immunomodulatory proteins (FIPs). These exhibit great potential towards inhibiting proliferation of hepatoma Hep G2 cells and breast cancer MCF-7 cells [73]. Furthermore there is no effect of thermal, freezing, acid, alkali and dehydration treatments on the properties of lectin protein (ABL) isolated from *A. bisporus*, hence indicating their usage as a stable immune stimulant for nutraceutical and functional food development [74]. Similar results have been reported in case of *G. lucidum* protein after giving various food processing treatments [75].

4.3. Terpenes

Terpenes are basically a group of volatile unsaturated hydrocarbons which are responsible for the anti-inflammatory activities and have been isolated from the mushrooms widely and these terpenoids are responsible for many pharmacological activities like anticancer antimalarial, anticholinesterase, antiviral, antibacterial and anti-inflammatory activities [76]. These are classified as monoterpenoids, sesquiterpenoids, diterpenoids and triterpenoids. Monoterpenes and sesquiterpenoids have been isolated

from the *Pleurotus cornicopiae* and the same have been investigated for their cytotoxicity against cancer line and found positive results [77]. Another species *Flammulina velutipes* has also been reported to contain sesquiterpenoids naming flammulinol, flammulinolides and were tested for their cytotoxicity against three tumor cell lines namely; HepG2, HeLa, and KB [78]. About the triterpenoids, lanostane is the compound which has been mostly isolated from the mushrooms, and reported as an anticancer agent. Two lanostane triterpenes isolated namely methyl ganoderate A acetonide and n-butyl ganoderate H from *Ganoderma lucidum* with anti-acetylcholinesterase activity [79]. Hence the usage of triterpenoids opened one more way to utilize them as a possible drug for the treatment of Alzheimer's and related neurodegenerative diseases by the pharmaceutical companies. Basically these terpenes can be effectively utilized in developing drugs for curing degenerative diseases. Some of the terpenes isolated from the different mushroom species along with the biological activities are given in Table 4 and their structures are given in Fig. 3.

4.4. Antioxidants

Physiological processes involve production of free radicals and other group of reactive oxygen as by-products [88]. Presence of free radicals beyond a limit in the body alters the antioxidant balance and results in oxidative stress. This leads to the destruction of tissue and cells which later develop into cancers. So to maintain the balance of the antioxidants in the humans' diets is an important factor to reduce theses oxidative damage. Phenolics, tocopherol, ascorbic acid and caretenoids are the responsible compounds which has been isolated from the different species of mushrooms and reported to boost the immune system, have anti-cancerous, anti-hypercholesterolaemic activity and anti-viral activity, and ameliorate the toxic effect of chemo- and radiotherapy [72]. Their radical scavenging activity has been extensively studied and documented, species like *Pleurotus* [89], *Agaricus* [90] *G. lucidum* [91] and *L.edodes* [92] known for their profound antioxidant activities. New Indian emerging medicinal mushroom species *P. rimosus* has been also studied for its antioxidant power, and found that its methanol and ethyl acetate extracts are good in lipid peroxidation, hydroxyl radical, and nitric oxide scavenging inhibiting activities [93]. The popular Chinese edible mushroom *Hericium erinaceus* is used to treat number of ailments, its hot water extract exhibit high free radical scavenging activity which strengthens its usage as nutraceuticals having natural antioxidants. The high phenolic contents and high antioxidant potential of *C. sinensis* makes it suitable for the healthy commercial preparations for the consumers [94].

4.5. Other biologically active nutraceutical compounds present in mushrooms

High utilization of medicinal mushrooms to prevent and treat the degenerative diseases has been observed worldwide. The medicinal implications of mushrooms include their role as an

Table 4
Terpenes and their activity in mushrooms [76].

Type of terpenes	Mushroom species	Compound	Activity
Sesquiterpinoids	*P. cornucopiae*	Pleurospiroketal	Cytotoxic
	F. velutipes	Enokipodin J	Cytotoxic
		2,5-Cuparadiene-1,4-dione	Antioxidant Antibacterial Cytotoxic Antioxidant Antibacterial
	F. velutipes	Flammulinolide	Cytotoxic Antibacterial
	F. velutipes	Enokipodin	Antimicrobial
	L. subpiperatus	Lactarolide A	Promotional NA
Diterpenoids	*P. eryngii*	Eryngiolide A	Cytotoxic
	H.erinaceum	Erinacine A	Antibacterial
	Tricholoma sp.	Tricholomalide A	Cytotoxic
		Tricholomalide B	
		Tricholomalide C	
Triterpenoids	*G. lucidum*	Methyl ganoderate A acetonide	Anticholinesterase
		n-Butyl ganoderate H	
		Methyl ganoderate A	
		Ganoderic acid B	
		Ganoderic acid E	
		Ganolucidic acid A	
		Ganodermadiol	
		Ganoderic acid Y	
		Ganoderiol F	
		Lucidumol B	
		Ganodermanondiol Ganodermanontriol	
		Lucidadiol	
	G. lucidum	Lucidenic acid N	Anti-invasive
		Lucidenic acid A	
	G. lucidum	*n*-Butyl lucidenate N	Effect of on adipocyte
		n-Butyl lucidenate A	differentiation in 3T3-L1 cells
	G. lucidum	Ganoderic acid Sz	Anticomplement Ganoderic acid C1
	G. amboinenese	Ganodermacetal	Toxic activity against
		Methyl ganoderate C	brine shrimp larvae
	G. lucidum	Ganoderic acid DM	Cytotoxic
	G. amboinenese	Ganoderic acid X	Cytotoxic

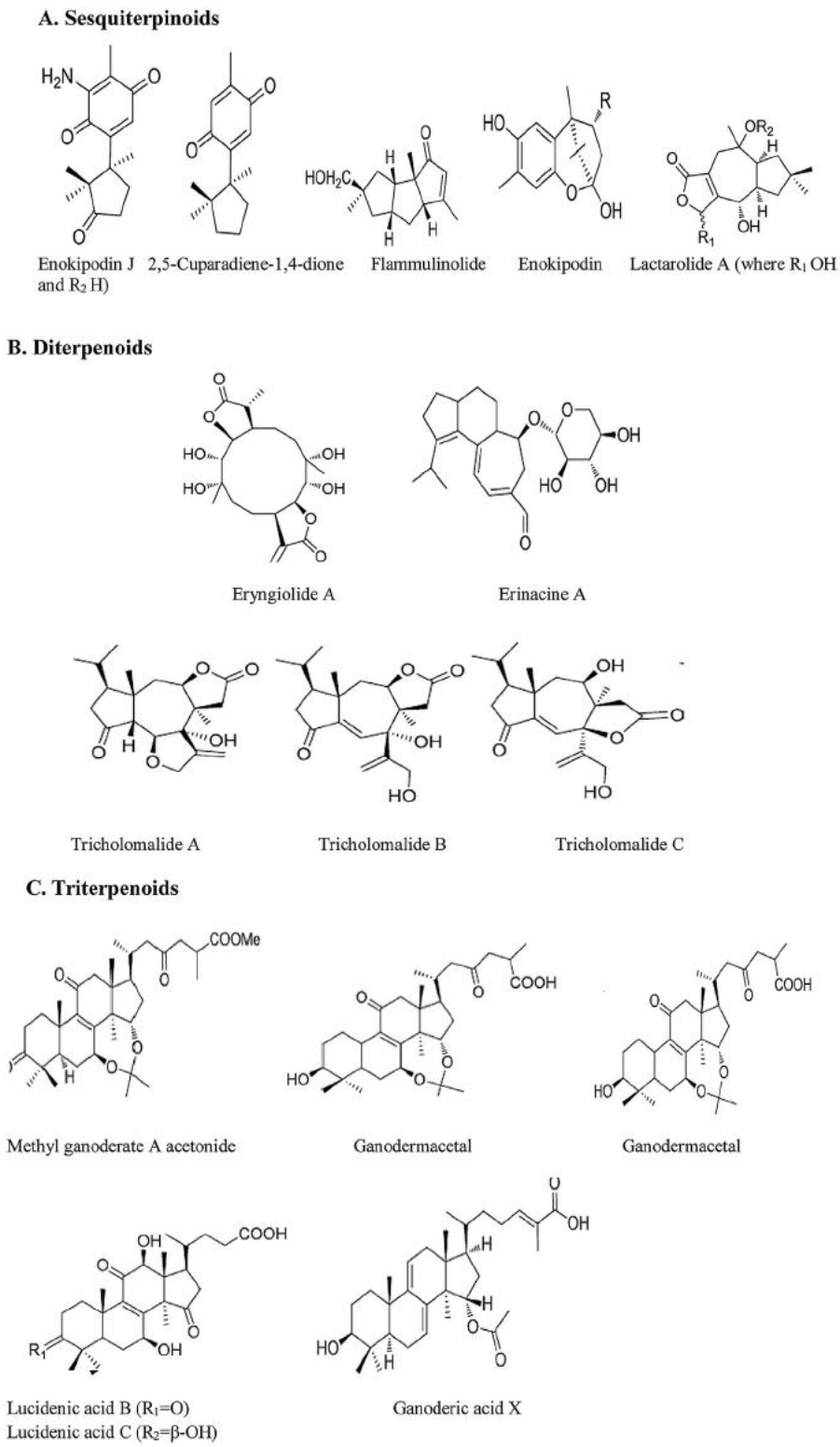

Fig. 3. Chemical structures of some bioactive terpenes [80–87].

antitumor, anticancer, antidiabetic, cardio-protector, hepato-protective, neuro-protective, antibacterial, antiviral etc. have been largely studied and documented. There are different biological compounds synthesized by these mushrooms which are responsible for the above written functions. Some of the identified molecules are β-glucan, dietary fibre, proteoglycan, lectin, phenolic compounds, flavonoids, volatile oils, tocopherols, phenolics, flavonoids, carotenoids, alkaloid, glycopeptides, folates and organic acids [95]. Caterpiller mushroom *Cordyceps militaris* contains a natural immunostimulating polysaccharide composed of glucose which extends its potential to cure various kind of cancers and tumours [96]. Moreover the tablet formulations made out of the medicinal mushrooms such as *Hericium erinaceus* has been extensively used for strengthening the gastrointestinal and

immune function. Further the medicinal effects of its bioactive compounds hericenones and erinancines are also known to prevent or treat human chronic, cognitive, and neurological diseases [97]. *Phellinus rimosus* again a superior edible fungus is known for its marvellous activity against tumour and for refreshing human body to improve longevity [98]. The isolated active molecules dictyophorine A and B from *Dictyophora indusiata* edible fungi accounts for curing neurodegenerative diseases by improving Nerve Growth Factor [73]. Japanese edible fungi *Panellus serotinus* (Mukitake) is thought to be helpful in preventing the development of nonalcoholic fatty liver disease [99]. Mushrooms have also emerged in the field of developing natural prebiotics. Literature supports that *Pleurotus* spp., *Lentinus edodes*, *Agaricus bisporus* and *Tremella fuciformis* possess strong prebiotic activities [100–102]. Some of the mushroom species along with their different bioactive molecules and therapeutic properties are given in Table 5.

4.6. Mushroom nutraceutical products

Mushrooms from last many decades are not being consumed as food but also utilized as major bioactive component in making of dietary supplements to improve the quality of human life. This reflects that the mushroom extracts are target specific and have been regularly tested in the clinical trials by the scientists and also by the pharmaceutical companies for developing functional foods. Because making such products from the fruit bodies, mycelia or its extracts have pharmacological benefits helping in the early intervention of sub-healthy states in humans which prevents the consequences of life threatening diseases [95]. Some commercially available nutraceuticals includes Organic ReiShi-Gen a synergistic blend of 50% *L. edodes* and 50% *G. lucidum* which is a daily health supplement providing immune strength. The list of other commercially available nutraceutical products along with the doses prescribed by the pharmaceutical companies is given in Table 7. However production of such nutraceuticals requires a deep knowledge about the structure and functional relationship, an economical viable, controllable production process and a stable polysaccharide quantity and quality [119]. Further it has been noticed that the doses used in the clinical studies vary subject to subject along with the ailment treated (Table 6); on the other hand the commercial products made from the various species are developed only to fulfil the overall daily nutritional requirements. In addition there is a strong urge of clinical trials to be run for these nutraceutical products in order to be accepted by the global market.

5. Future scope

A lot of compiled data is available on the classification, structure, biology, nutritional composition, post harvest technologies and value addition of different species of mushrooms. But extensive research is required in the field of genomics, proteomics, secretion of secondary metabolites and their involvement in biochemical pathways governing their therapeutic roles. The exact mechanisms involved in these therapeutic actions should be investigated through new advance approaches for the clear understanding to the common people. The role of β-glucans in addressing several diseases is not new to us [116]. But the active mechanism involved, the role of structure and molecular weight behind that is still unclear as all kind of β-glucans are not medicinally active [117]. Some areas like biochemistry of medicinally active polysaccharide–peptide complexes involved in therapeutic activities require more attention [62]. Further as per today's scenario the acceptability of fresh mushrooms in the market is more. Due to the lack of proper information and awareness consumers are least interested in the dietary supplementations developed using medicinal mushrooms. The research on the dietary supplements, their biochemical actions, reactions and interaction studies with other drugs is lacking. Mushrooms are traditional medicines which are ignored today due to the lack of maintaining the proper standards during manufacturing, the purity system, and insufficient clinical trials [118]. So for the future our main emphasis should be to create awareness amongst the consumers regarding the proper utilization of this golden drug. This can only be obtained by opting the appropriate standards and practices for making the drugs out of them and should be compared with the Western and Eastern medicines.

Table 5
Other bioactive compounds isolated from mushrooms [103].

Mushroom Species	Bio active Molecules	Medicinal Properties
Agaricus bisporous	Lectins	Enhance insulin secretion, anti-aging property.
Auricularia auricula	Acidic Polysaccharides	Anti-tumour activities, lowers cholesterol, triglycerides, and lipid levels; decrease blood glucose, beneficial in coronary heart disease, immune tonic
Cordyceps sinensis	Cordycepin	Cure lung infections, hypoglycemic activity, cellular health properties, anti-depressant activity.
Flammulina velutipes	Polysaccharide, flammulin, FVP (*Flammulina* polysaccharide protein), peptide glycans, prolamin (active sugar protein), Proflamin (glycoprotein)	Antioxidant, anti-cancer activity, anti-ageing property; immuno-modulatory, anti-viral action.
Ganoderma lucidum	Polysaccharides, triterpenoids, germanium, nucleotides and nucleosides, Ganoderic acid, Beta-glucan,	Augments immune system, liver protection, antibiotic properties, inhibits cholesterol synthesis; immunomodulatory, anti-cancerous properties.
Grifola frondosa	Polysaccharide, Lectins	Increases insulin secretion, decrease blood glucose, improves ovulation.
Lentinula edodes	Eritadenine, Lentinan	Lower cholesterol, anti-cancer agent.
P. florida		anti-hyperglycaemic; anti-hypercholesterolemia effect
P.sajor-caju	Lovastatin polysaccharide	Lower cholesterol, prevents cardiovascular disorders.
Trametes versicolor	Polysaccharide-K (Kresin), Coriolon and glycoproteins	Decrease immune system depression, prevents cancer, inhibits growth of *Candida albicans*, anti-viral activity by inhibiting the replication of HIV, liver protective functions.
Volvariella volvacea	Glycoproteins	Cardio-protective, lowers blood pressure.
Hericium erinaceus	Hericenones and erinacines	Neuritogenic effects

Table 6
Clinical studies on mushrooms conducted from 2013–2016.

Species name	Ailment treated	Dosage	Subjects	Duration	Outcomes	References
A. blazie	Multiple myeloma	AndoSan extract, containing 82% of *A. blazei* Murill	40 patients with multiple myeloma, treatment (n = 19), control (n = 21)	7 weeks	. Physiological Effects: No statistically significant clinical impact of AndoSan was detected	Tangen et al. [104]
	Cancer remission	Senseiro, Murill granulated powder; 1 packet = 1800 mg	67 cancer patients; group 1 (n = 23), group 2 (n = 22), group 3: (n = 22) group1- 1 pack/day; group2- 2 packs/day; group-3 3 packs/day	6 months	Cancer remission- Males improved more in physical components, while females improved only mental components. Optimal dose effects occurred at 2 packs/day	Ohno et al. [105]
	Oxidative and functional parameters of the brain tissue and brain mitochondria	200 mg/kg	Rats	21 days	*A. blazei* extract was protective to the brain against oxidative stress by decreasing the lipid peroxidation levels and the reactive oxygen species content and by increasing the nonenzymic and enzymic antioxidant capacities	De Sá-Nakanishi et al. [106]
A. brasiliensis	Hypertension	10%	Hypertensive rats	5 weeks	Improved hypertensive cardiovascular hemodynamics & lowered sympathetic nervous tone in rats were observed	Tsubone et al. [107]
	HIV	30 mg/kg of body weight/day	24 children (2–8 yrs old)	3 months	Supplementation of diet with *A. sylvaticus* reversed the oxidative alterations and improved antioxidant defense in infected individuals	Figueira et al. [108]
Flammulina velutipes	Hypolipidemic	1, 2, and 3% *F. velutipes* Powder (FVP)	Male hamsters	8 weeks	3% FVP have the lowest concentration of TC (total cholesterol), TG (triacylglycerol), LDL (low density lipoprotein cholesterol), and LDL/HDL (high density lipoprotein cholesterol) in the serum and liver	Yeh et al. [109]
P. ostreatus and *P. cystidiosus*	Diabetes	50 mg/kg/body weight	Type 2 diabetic patients	2 weeks	Increased utilization of glucose by peripheral tissues, reduced glycogen synthase kinase with improved glycogen synthesis	Jayasuriya et al. [110]
P. ostreatus	Diabetes	3gm powder capsule	27 hypertensive male type 2 diabetes mellitus (age range 32 to 68 years)	3 months	Decreased fasting plasma glucose level indicated no detrimental effect on renal system	Choudhury et al. [111]
Pleurotus ferulae	Diabetes	250 mg/kg body weight/day	Streptozotocin-induced diabetic rats	6 weeks	Insulin levels increased and HbA1c levels decreased	Wang et al. [112]
Lentinus edodes	Hyper-lipidemia	Diet containing 10% *L. edodes* mushrooms	20 males and 20 females rats	3 months	62% reduction of serum leptin concentrations in females and increased serum insulin (137%) and decreased serum glucose (15%) in males were observed	Shanggong et al. [113]
G. lucidum	Diabetes	Spores powder 1 g/day	Type 2 diabetic rats	4 weeks	Promoted glycogen synthesis and inhibited gluconeogenesis.	Wang et al. [114]

Table 7
List of some commercial products available in the market [115].

Mushroom species	Products in the market	Doses recommended	Health benefit
G. lucidum	Pure Red Reishi −capsules Organic Reishi tablets – MRL GanoSuper-Concentrated extracts for Coffee	1capsule/day (Each containing 500 mg of pure Ganoderma) 3 capsule/day (Each containing 500 mg of pure Ganoderma) 2capsules/day (500 mg containing 37% Polysaccharides and 4.6% Triterpines active ingredients)	Increase the body's resistance to stress and help it overcome all health challenges more quickly Supports the body's normal cellular immune system
50% G. lucidum and *50% L. edodes*	Organic ReiShi-Gen (synergestic formulations)	–	Immune support supplement
A. blazei Murill	*Agaricus blazie* capsules	1capsule/day (Each containing 400 mg fruit body of *A. balzie*)	Immune support supplement.
L. edodes	Shiitake Gold-capsules	3 capsules/day (500 mg containing pure containing pure shiitake)	For complete physical health
Cordyceps & Ganoderma	Nutricafe-Organic Performance Coffee	–	Increases physical endurance and helps to remove toxins from your body.
Blend of *C. sinensis, A.blazei, L. edodes, Grifola frondosa, G. lucidum, and Coriolus versicolor)*	Immune Assist-complete	2 capsule/day (375 mg of pre-Fukushima, 50 mg of soluble 1,3–1,6 Beta Glucans and 125 mg of pure immune-active Heteropolysaccharides)	Immunity supporter
Blend of *(Agaricus, Cordyceps, Ganoderma and L. edodes)*	Cocoa Mojo- cocoa powder	–	Immunity supporter

6. Conclusion

Due to the diverse habitat and varied ecological zones across the world, varieties of mushrooms emerge as the next generation's nutraceutical food. Mushrooms are considered to be a complete health food and suitable for all age groups due its rich protein, dietary fiber, vitamins and mineral contents. Along with the immense nutritional supremacy they also knew to contain various bioactive components which are being continuously used as a raw material for developing functional foods by the pharmacists. Such bioactive molecules are polysaccharides, terpenoids, low molecular weight proteins, glycoproteins, and antioxidants etc., which have a great role to play in boosting immune strength, lowering risks of cancers, inhibition of tumoral growth, blood sugar maintenance and much more. The retention of these bioactive molecules needs to be trapped in such a way that they remain intact from farm to the fork level, so the losses due to the various processes like cooking, canning, blanching and heating should be in such a manner that minimum losses occur. Also, some species contains anti-nutritional components that may harm the human health, but further it needs more scientific reports to establish the fact related to the kind of species that may harm and the state of consuming them (raw or cooked). So overall we can say that the spectrum of various physiological activities governs by the mushrooms are wide and it is anticipated that with the current science and advancements in molecular biology and nutrigenomics mushrooms can emerge as one of the best nutraceutical food. This is only possible if we have integrated information regarding all the aspects of nutraceutical properties for its better utilization in development of functional foods.

Acknowledgements

The authors are highly grateful to University Grant commission (UGC) and National Horticulture Board (NHB), India for providing financial assistance.

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